

Erdős Problem #588

EP-588

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $f_k(n)$ be minimal such that if n points in $\mathbb{R}^2$ have no $k + 1$ points on a line then there must be at most $f_k(n)$ many lines containing at least k points. Is it true that $f_k(n) = o(n^2)$ for $k \geq 4$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>